Draft version October 29, 2026
Typeset using LaTeX **twocolumn** style in AASTeX7.0.1

Precovery of the Dwarf Planet Candidate 2026 ED_{138} in 2010–2011 VISTA/VVV Images

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ABSTRACT

2026 ED_{138} is a large trans-Neptunian object and dwarf planet candidate ($a = 42.9$ au, $H = 2.93$) reported by the Rubin Observatory in 2026 and linked to Dark Energy Camera images back to 2016. I searched public survey archives for earlier images. In 2010–2011 the object crossed the Galactic plane inside the footprint of the VISTA Variables in the Vía Láctea (VVV) survey. Using template-subtracted pawprint stacks and a local astrometric tie to Gaia DR3, I measured 15 positions on four nights (2010 March 18 to 2011 July 1) in the Z, Y, J and H bands. They agree with the current orbit to an rms of 0.06″ in right ascension and 0.02″ in declination, and extend the observed arc by six years. The positions have been submitted to the Minor Planet Center. I also report searches that found nothing: Catalina Sky Survey, ATLAS, SkyMapper, Gaia DR1–DR3 and VVV K_s images.

Keywords: Trans-Neptunian objects (1705), Astrometry (80), Sky surveys (1464)

1. INTRODUCTION

2026 ED_{138} was first reported from the Simonyi Survey Telescope of the Rubin Observatory (code X05) on 2026 March 4. The orbit in MPEC 2026-U85 (Minor Planet Center 2026) adds 44 Dark Energy Camera (DECam, W84) positions from 2016 to 2023, for 74 observations over 6 oppositions ($a = 42.95$ au, $e = 0.142$, $i = 24.5°$, $H = 2.93$). With $H = 2.93$ the diameter is roughly 750–1100 km for geometric albedos of 0.1–0.2 (P. Pravec & A. W. Harris 2007), large enough that it is probably in hydrostatic equilibrium, which makes it a dwarf planet candidate; its albedo and shape have not been measured. During 2009–2023 the object stayed near $\delta \approx -45°$ and moved from Galactic latitude $b \approx +3°$ to $-9°$ at $l \approx 338°$–$349°$, through crowded fields towards the bulge. This note reports a search of public archives for observations before 2016.

2. ARCHIVE SEARCH AND NULL RESULTS

I used the Planetary Data System CATCH service[2] and CADC SSOIS (S. D. J. Gwyn et al. 2012) with JPL Horizons (J. D. Giorgini et al. 1996) ephemerides. The expected brightness is $V \approx 19.7$; extrapolating the DECam colors gives rough estimates of $J \approx 18.1$ and $K_s \approx 17.2$ (Vega).

Catalina Sky Survey. CATCH returned no Catalina frames. From Mt. Lemmon the object never rises above about 13°, and the southern Siding Spring survey ended in 2013 and is not in CATCH.

ATLAS (J. L. Tonry et al. 2018). 2666 frames contain the predicted position, with median 5σ limits of 18.5–18.8 and none deep enough to reach $V = 19.7$.

SkyMapper DR4 (C. A. Onken et al. 2024). Of 41 frames, I measured the eight usable 100 s exposures from 2019–2020 by forced photometry. Crowding limits them to 5σ depths of 17.3–18.5, giving an expected signal-to-noise ratio (S/N) of 0.9–1.9; the largest measured value was 2.7, 1.7″ from the prediction, which I treat as noise. Two 2019 August 15 frames were unusable because the object fell at the cutout edge.

Gaia DR1–DR3 (Gaia Collaboration et al. 2016, 2023). The object is not in the DR3 solar-system tables, which contain numbered objects only. I counted faint ($18.5 < G < 21.5$) two-parameter sources within 2″ of the track during each release's observing window and compared the counts with tracks shifted by ±40″ and ±80″ in declination. The real track has no excess (DR1: 622 vs. a control mean of 608; DR2: 1416 vs. 1415; DR3: 1949 vs. 1911). This is expected: the object moves about 4″ between Gaia's paired field-of-view transits, so its detections are never merged into a published source.

VVV K_s. SSOIS lists 831 K_s exposures from 2010–2014. Forced photometry of 135 pawprint stacks on 37 nights gave a median S/N of 0.38 times the expected value, so the object is fainter in K_s than estimated. Fourteen stacks on seven nights reached S/N ≥ 5, but

[2] https://catch.astro.umd.edu

those that could be fitted lie 0.2″–1.4″ from the prediction, inconsistent with the < 0.1″ agreement in the other bands; I attribute them to blends or subtraction residuals and do not report them.

3. VVV PRECOVERY

SSOIS found the object in VVV (D. Minniti et al. 2010) images taken with VISTA (W. Sutherland et al. 2015) in 2010–2011, when it crossed the Galactic plane inside the VVV disk area. I retrieved 1′ cutouts of pawprint stacks from the VISTA Science Archive (DR5; N. J. G. Cross et al. 2012). For each science stack, stacks of the same field from other nights, when the object was at least 30″ away, were resampled onto the science pixel grid, scaled to its photometric zero point and median-combined into a template. I fitted a two-dimensional Gaussian to the template-subtracted image at the predicted position. The pipeline astrometry was then retied locally to Gaia DR3 using stars with $13 < G < 19$ in each cutout, corrected for proper motion to the epoch of observation. Mid-exposure times come from the detector time stamps (accurate to about ±10 s, or 0.01″ of motion).

The object is detected on four nights at S/N 6.6–30 (Figure 1), consistent with its expected brightness in each band. I excluded the three H-band stacks of 2010 March 25, where a nearby bright star shifts the centroid by 0.2″–0.3″, and no stacks were returned for 2010 March 27. The 15 remaining positions (Figure 2) have residuals from the JPL orbit of mean $(-0.055'', +0.008'')$ and rms $(0.060'', 0.021'')$ in $(\Delta\alpha\cos\delta, \Delta\delta)$, all within 0.11″. The JPL 3σ prediction uncertainty at these epochs is about $2.5'' \times 0.45''$, so the positions fall well inside it and will shrink it. The small RA offset is common to all four nights and probably reflects the current orbit.

The positions were submitted to the Minor Planet Center in ADES format on 2026 October 28 (observatory code W91, submission ID 2026-10-28T16:38:07.859_0000CHyD). At the time of writing they have been received but not yet accepted. Once included, the observed arc will run from 2010 to 2026. A longer arc will improve occultation predictions, which, together with thermal measurements and a rotational light curve, can establish the object's size and shape.

DATA AVAILABILITY

The positions (ADES PSV), photometry tables, analysis code and figures are archived with this note on Zenodo (https://doi.org/10.5281/zenodo.23740113). Code is MIT-licensed; text and figures are CC BY 4.0.

USE OF AI TOOLS

This work was done with substantial help from an AI assistant, Claude (Anthropic; model Claude Opus 5.5, used through Claude Code). Under the author's direction, the assistant ran the archive queries, wrote and ran the analysis code, made the measurements and figures, and drafted the text. The author chose what to pursue, reviewed the results and the manuscript, made the MPC submission, and is responsible for the content.

ACKNOWLEDGMENTS

Based on data products from VVV observations made with the VISTA telescope at the ESO Paranal Observatory under programme ID 179.B-2002, retrieved from the VISTA Science Archive (WFAU). This work used the Solar System Object Image Search of the Canadian Astronomy Data Centre, the PDS Small Bodies Node CATCH service, JPL Horizons, the Minor Planet Center, SkyMapper and ATLAS data, and data from the ESA mission Gaia, processed by the Gaia Data Processing and Analysis Consortium.

Software: Astropy (Astropy Collaboration et al. 2022), NumPy, SciPy, Matplotlib

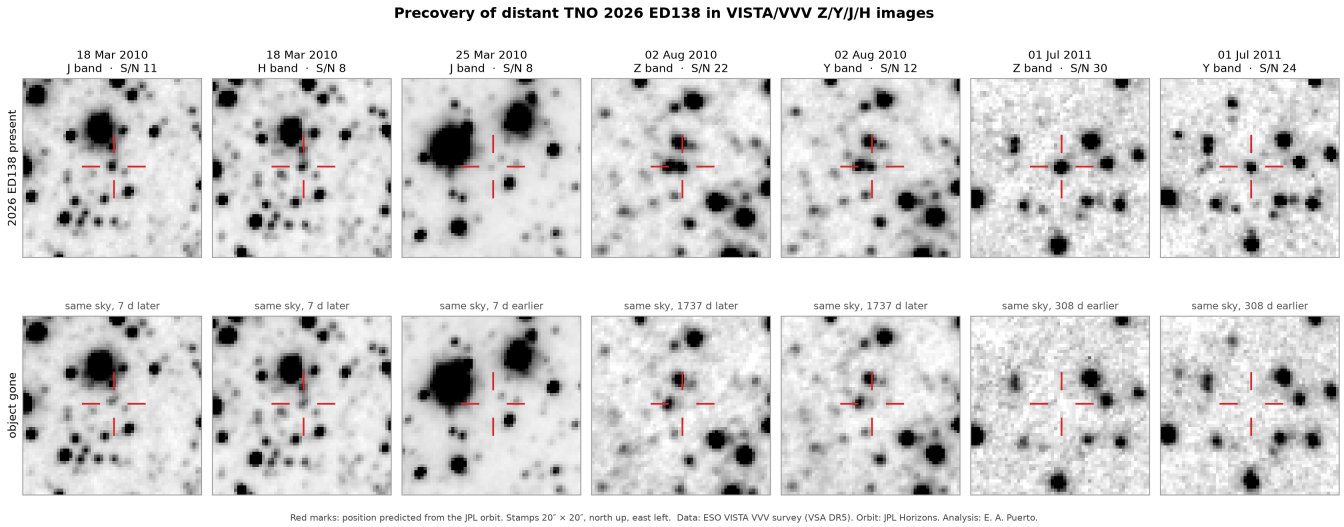


Figure 1. VVV detections of 2026 ED_{138}: the best stack for each night and band (top) and the same sky on another night (bottom). Red marks show the JPL-predicted position. All 15 frames are in the data archive.

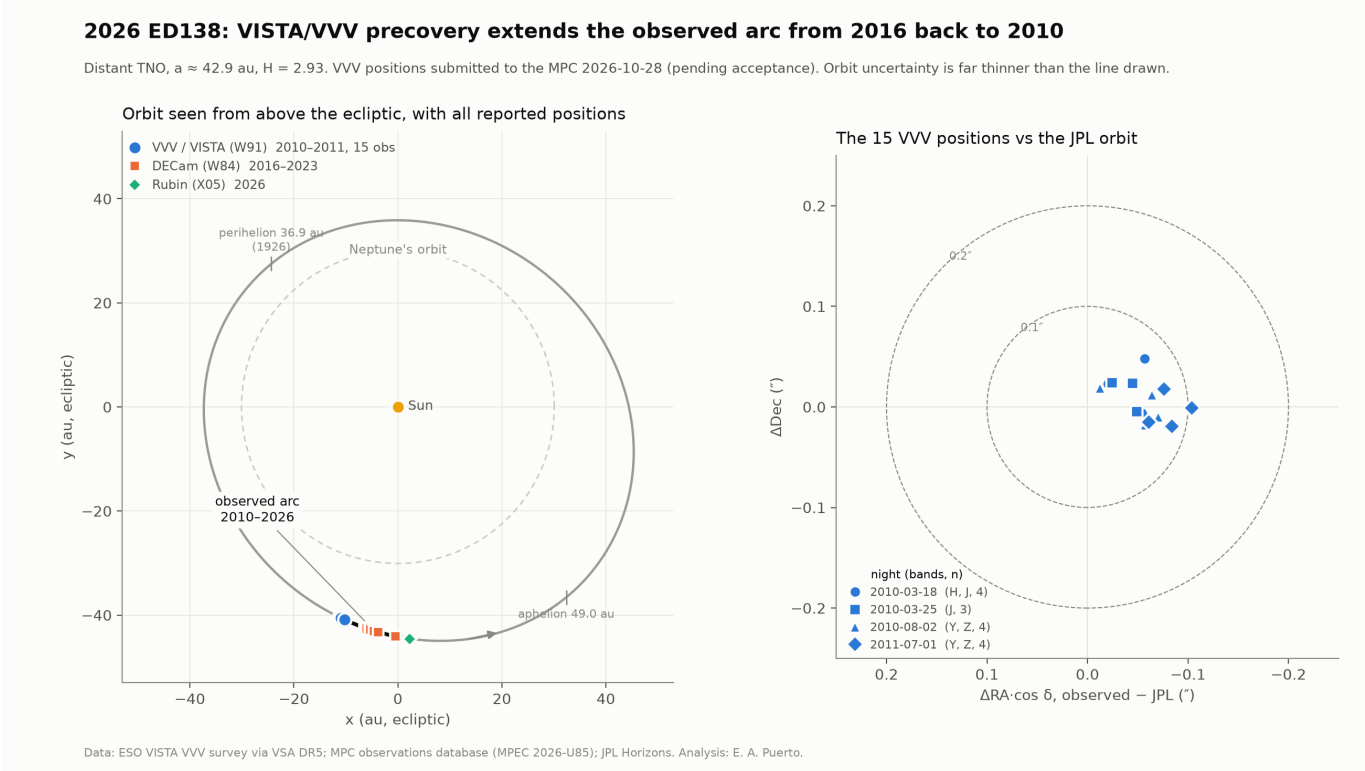


Figure 2. Left: the orbit seen from above the ecliptic, with the VVV (2010–2011), DECam (2016–2023) and Rubin (2026) positions. Right: residuals of the 15 VVV positions from the JPL orbit.

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